Practical 5

Program –

```
#include <SoftwareSerial.h>

// HC-05 connections
// Arduino RX = Pin 10
// Arduino TX = Pin 11
SoftwareSerial BT(10, 11);

const int LED_PIN = 8;

String password = "2468";
String input = "";

void setup() {
  pinMode(LED_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);

  Serial.begin(9600);
  BT.begin(9600);

  Serial.println("Bluetooth Password Control");
  Serial.println("Type password in mobile app");
  Serial.println("Correct Password: 2468#");
  Serial.println("--------------------------------");

  BT.println("Enter password followed by #");
}

void loop() {

  while (BT.available() > 0) {

    char ch = BT.read();

    // Ignore Enter and newline characters
    if (ch == '\r' || ch == '\n') {
      continue;
    }

    // Store password characters
    if (ch != '#') {

      input += ch;
      Serial.print(ch);
    }
```

```
    // Check password when '#' is received
    else {

      Serial.println();
      Serial.println("Checking password...");

      if (input == password) {

        digitalWrite(LED_PIN, HIGH);

        Serial.println("Access Granted!");
        Serial.println("LED STATUS: ON");

        BT.println("Correct Password - LED ON");
      }

      else {

        digitalWrite(LED_PIN, LOW);

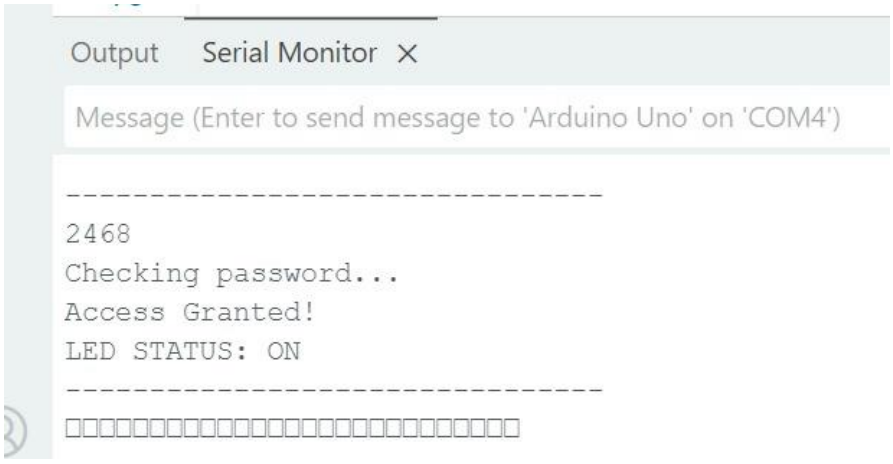
        Serial.println("Access Denied!");
        Serial.println("LED STATUS: OFF");

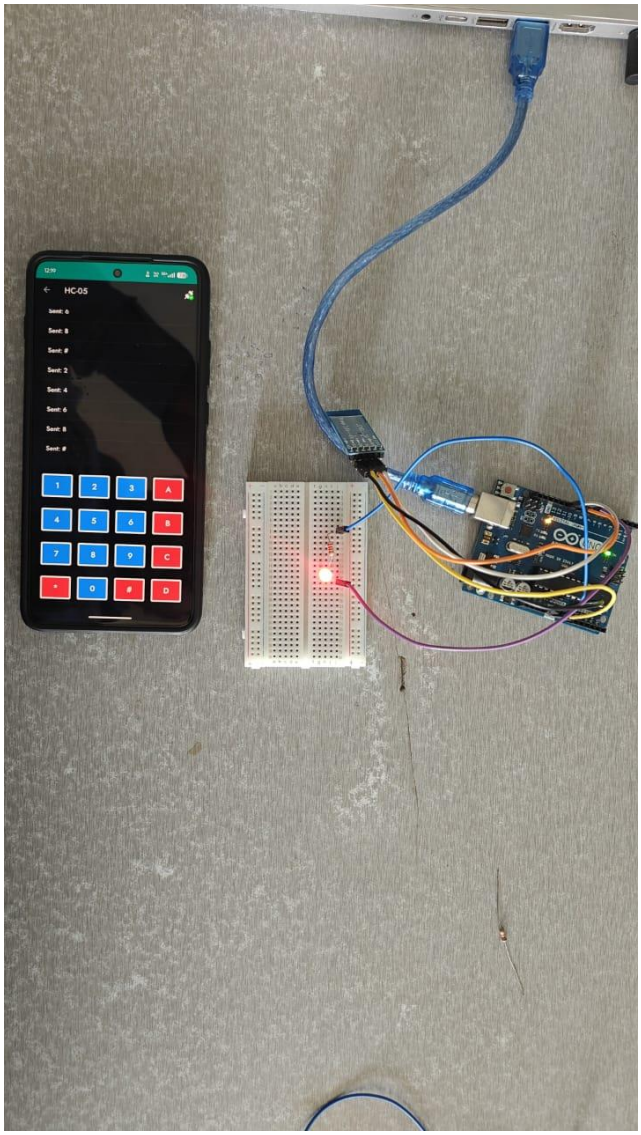
        BT.println("Wrong Password - LED OFF");
      }

      Serial.println("--------------------------------");

      // Clear input for next attempt
      input = "";
    }
  }
}
```

Output –




HC-05
Sent: 6
Sent: 8
Sent: #
Sent: 2
Sent: 4
Sent: 6
Sent: 8
Sent: #
1
2
3
A
4
5
6
B
7
8
9
C
*
0
#
D